HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained **3D BIM models** and technical drawings for infrastructure projects using Autodesk Revit, adhering to project specifications and BIM standards.
- Supported multidisciplinary design coordination efforts by performing **clash detection** using Navisworks, identifying and documenting over **150 clashes** across architectural, structural, and MEP disciplines.
- Assisted in the production of project documentation, including **BIM execution plans** and digital deliverable reports, ensuring compliance with contractual requirements.
- Applied **BIM standards** and workflows, including data management and quality assurance procedures, to ensure model integrity and consistency across project lifecycles.
- Collaborated with engineers, architects, and digital specialists in **multidisciplinary teams** to integrate design changes and deliver efficient digital solutions.
- Supported the adoption of **digital engineering best practices** by participating in training sessions and providing feedback on new workflows and software implementations.
- Containerized core BIM processing tools using **Docker** for consistent deployment across development and testing environments on **AWS EC2** instances.

SKILLS

BIM Software – Autodesk Revit, Navisworks, AutoCAD

Digital Engineering – Building Information Modeling (BIM), 3D Modeling, Clash Detection, Digital Deliverables, BIM Standards, Quality Assurance

Programming & Scripting – Python, Shell Scripting

Cloud & DevOps – AWS (EC2, S3, CloudWatch), Docker, Git, Jenkins

Databases – SQL, PostgreSQL

Operating Systems – Linux, Windows

PROJECTS

Infrastructure Project Model Development (HCL Technologies)

- Developed and refined **3D BIM models** for a large-scale regional infrastructure project, integrating architectural, structural, and MEP elements.
- Executed automated **clash detection** routines in Navisworks, reducing manual review time by approximately **30%** and flagging over **200 potential interferences**.
- Managed model data integrity and adherence to project-specific **BIM standards**, ensuring consistent data quality for downstream analysis and reporting.

Automated Report Generation for Design Deliverables (HCL Technologies)

- Created **Python scripts** to automate the generation of standardized digital deliverables and reports from BIM models, saving an estimated **10 hours** of manual effort per project phase.
- Integrated script execution into **Jenkins CI/CD pipelines** for repeatable and scheduled report generation, improving delivery timelines for client documentation.
- Stored generated reports and model backups in **AWS S3**, establishing a centralized and accessible repository for project artifacts.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)